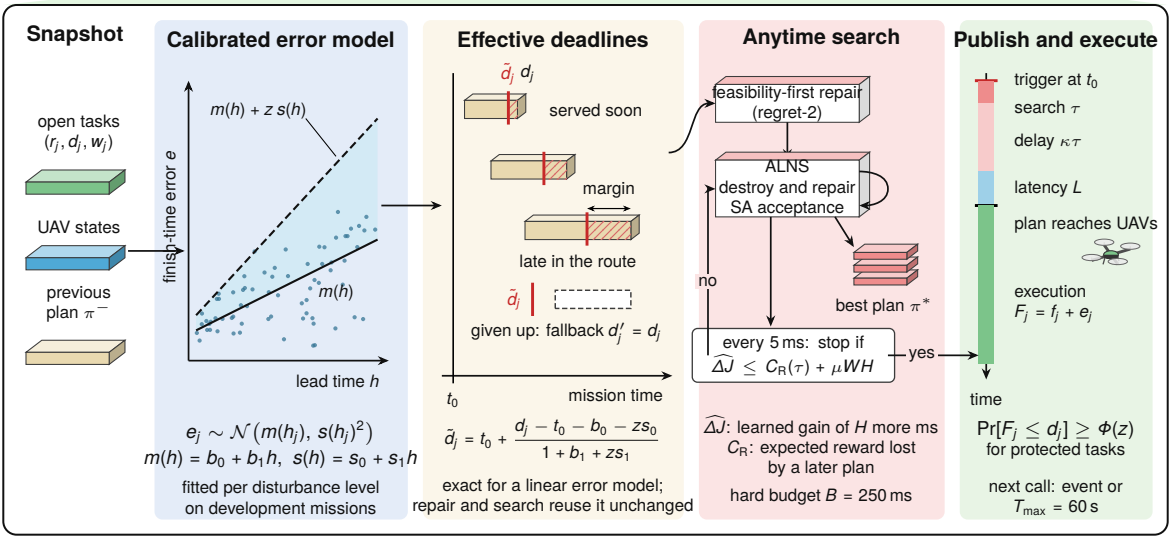

telemetry: UAV states, actual finish times F_j, failures
wind
event
AnytimeSwarm-CC
plan π^*
Utility, late tasks
Mission
Snapshot at t_0
Replanner (detail below)
Tier B or tier C (PX4)


Snapshot
open tasks (r_j, d_j, w_j)
UAV states
previous plan π^-
Calibrated error model
$m(h) + z\,s(h)$
finish-time error e
$m(h)$
lead time h
$e_j \sim \mathcal{N}\left(m(h_j),\, s(h_j)^2\right)$
$m(h) = b_0 + b_1 h,\ s(h) = s_0 + s_1 h$
fitted per disturbance level on development missions
Effective deadlines
$\tilde{d}_j$ d_j
served soon
margin
late in the route
$\tilde{d}_j$
given up: fallback $d'_j = d_j$
t_0
mission time
$\tilde{d}_j = t_0 + \dfrac{d_j - t_0 - b_0 - z s_0}{1 + b_1 + z s_1}$
exact for a linear error model; repair and search reuse it unchanged
Anytime search
feasibility-first repair (regret-2)
ALNS destroy and repair SA acceptance
best plan π^*
no
every 5 ms: stop if $\widehat{\Delta J} \le C_R(\tau) + \mu WH$
yes
$\widehat{\Delta J}$: learned gain of H more ms
C_R: expected reward lost by a later plan
hard budget B = 250 ms
Publish and execute
trigger at t_0
search τ
delay $\kappa\tau$
latency L
plan reaches UAVs
execution $F_j = f_j + e_j$
time
$\Pr[F_j \le d_j] \ge \Phi(z)$ for protected tasks
next call: event or T_{max} = 60 s